

Impact of Back-to-Base Normothermic Machine Perfusion on Complications and Costs

A Multicenter, Real-World Risk-Matched Analysis

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Objective: Assess cost and complication outcomes after liver transplantation (LT) using normothermic machine perfusion (NMP).

Background: End-ischemic NMP is often used to aid logistics, yet its impact on outcomes after LT remains unclear, as does its true impact on costs associated with transplantation.

Methods: Deceased donor liver recipients at 2 centers (January 1, 2019, to June 30, 2023) were included. Retransplants, splits, and combined grafts were excluded. End-ischemic NMP (OrganOx-Metra) was implemented in October 2022 for extended-criteria donation after brain death (DBDs), all donations after circulatory deaths (DCDs), and logistics. NMP cases were matched 1:2 with static cold storage controls (SCS) using the Balance-of-Risk [donation after brain death (DBD)-grafts] and UK-DCD Score (DCD-grafts).

Results: Overall, 803 transplantations were included, 174 (21.7%) receiving NMP. Matching was achieved between 118 NMP-DBDs with 236 SCS; and 37 NMP-DCD with 74 corresponding SCS. For both graft types, median inpatient comprehensive complications index values were comparable between groups. DCD-NMP grafts experienced reduced cumulative 90-day comprehensive complications index (27.6 vs 41.9, $P = 0.028$). NMP also reduced the need for early relaparotomy and renal replacement therapy, with subsequently less frequent major complications (Clavien-Dindo $\geq$ IVa). This effect was more pronounced in DCD transplants. NMP had no protective effect on early biliary complications. Organ acquisition/

preservation costs were higher with NMP, yet NMP-treated grafts had lower 90-day pretransplant costs in the context of shorter waiting list times. Overall costs were comparable for both cohorts.

Conclusions: This is the first risk-adjusted outcome and cost analysis comparing NMP and SCS. In addition to logistical benefits, NMP was associated with a reduction in relaparotomy and bleeding in DBD grafts, and overall complications and post-LT renal replacement for DCDs. While organ acquisition/preservation was more costly with NMP, overall 90-day health care costs-per-transplantation were comparable.

Keywords: liver transplantation: normothermic machine perfusion, complications, costs, health care costs

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End-stage liver disease (ESLD) affects 5.2 million patients annually and is increasing in incidence and prevalence in the United States and globally.^{1–3} The only definitive treatment for ESLD providing long-term survival is liver transplantation (LT).^{4,5} Thus, the volume of LT continues to rise annually, with over 9000 transplants performed in 2021 in the United States.⁶ While post-transplant outcomes are improving, there remains both a critical lack of access to life-saving liver grafts and a higher than desirable rate of serious post-transplant complications.^{7–9}

Ex-situ machine perfusion is rapidly becoming an essential component of LT. Both normothermic (NMP) and hypothermic-oxygenated (HOPE) machine perfusion demonstrate profound potential at this time, with touted benefits including improved post-transplant graft function, reduction of overall and biliary complications, and increased organ utilization, specifically regarding marginal grafts.^{10–17} However, the true clinical impact of NMP, which is currently the only food and drug administration-approved technique in the United States, remains unclear, especially in the US, where machine perfusion is newer to clinical practice. Machine perfusion is also supposedly quite costly. For example, while there are no real-world US-based cost-analyses, a recent Canadian study reports per-use costs of nearly \$50,000 per NMP-run.¹⁸

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Since the approval of NMP in the United States in late 2021, no study has systematically compared the outcomes of end-ischemic NMP-treated grafts versus those treated with standard static cold storage (SCS). In addition, there has not been a primary reporting of costs associated with cumulative complications after NMP in risk-matched cohorts. We provide here the first report of complications and costs within 90 days of LT in a risk-matched analysis between grafts treated with NMP and SCS.

METHODS

This retrospective cohort analysis included patients undergoing primary deceased donor LT from January 1, 2019, to September 30, 2023, at 2 academic transplant programs (Cleveland Clinic, Cleveland, OH, and Cleveland Clinic Florida, Weston, FL). Patients were identified from a prospectively maintained institutional database. Only patients with at least 90 days of active follow-up were included. Living donor LT, combined transplant, retransplantation, split grafts, or domino transplants were excluded. The study was approved by the Institutional Review Board (IRB No. 23-1348 and 24-241) and was conducted in accordance with the Declaration of Helsinki.

Routine use of end-ischemic ex situ NMP (OrganOx *Metra*) began in our centers on October 1, 2022. Programmatic guidelines for machine perfusion are listed in Supplemental Table 1, <http://links.lww.com/SLA/F63>; generally, this includes the use of NMP for all donation after circulatory death (DCD) grafts, extended-criteria or otherwise risky donation after brain death (DBD) grafts, and for logistical reasons. However, the final decision to employ NMP is made at the discretion of the accepting transplant surgeon. In cases of machine perfusion, liver allografts were perfused using a back-to-base approach, wherein the organ is procured as per national standard with cold flush and SCS, followed by transport to the accepting recipient center. After back table preparation, the liver is connected to OrganOx *Metra* device and perfused until the recipient hepatectomy is completed. Both centers aim for at least 4 hours of NMP to assess liver graft viability, using lactate clearance and bile chemistry analysis prior to the decision to transplant the organ.¹⁹

Outcomes

The electronic medical record was reviewed for donor and recipient information. Data were supplemented from the United Network for Organ Sharing and Scientific Registry of Transplant Recipients databases where necessary. The overall risk of the transplant, including both donor/graft and recipient risk, was estimated using the previously validated Balance-of-Risk (BAR) score for DBD grafts^{20–22} and UK-DCD score for DCD grafts.²³ The primary outcomes of the study were the median Comprehensive Complications Index (CCI) and the costs associated with transplantation. The CCI was calculated for the transplant admission and the total 90-day postoperative period as the summation of the Clavien-Dindo graded complications over that period as previously described and validated.^{24–27} Secondary outcomes included rates of specific complications up to 90 days post-LT, including new requirement of renal replacement therapy (RRT), rate of relaparotomy, vascular complications, bleeding requiring intervention, biliary complications including anastomotic stricture (AS), nonanastomotic stricture (NAS), bile leaks,

acute cellular rejection, 90-day graft loss, and 90-day mortality. Complications were graded using the validated descriptions in accordance with the recent Delphi consensus, as published by Gero et al and Muller et al.^{28,29}

Cost data were obtained for each patient for the time before and after LT, including 90 days before transplant (90-day pre-LT), inpatient costs (including the costs of surgical and postoperative care), the total 90-day period following LT (including the admission costs; 90-day post-LT), and the total period from 90 days pre-LT to 90 days post-LT (90-day pre-/post-LT). The costs of machine perfusion, including disposables, are included in the “Organ Acquisition/Preservation” category. As these costs are expected to be higher in the NMP cohort, the specific cost of postoperative care, excluding organ acquisition/preservation was also reported. Cost data in all periods reflect costs incurred within the included health care systems and thus excludes care received at other institutions during these periods.

Statistical Analysis

The primary comparison of interest was NMP versus SCS. As there are known differences in risk profile between DBD and DCD grafts, a risk-matched analysis was performed for DBD grafts (DBD-NMP vs DBD-SCS) and separately for DCD-grafts (DCD-NMP vs DCD-SCS). NMP-treated grafts (intervention) were matched in 1:2 manner to their SCS counterparts (control) using the BAR and UK-DCD scores for DBD and DCD grafts, respectively.

Descriptive statistics were used to summarize the study population. Continuous variables were represented as median and interquartile range (IQR). Categorical variables were represented as counts and percentages. Categorical variables were analyzed using chi-square tests or Fisher exact tests as appropriate. Continuous variables were analyzed using *t* tests or Mann-Whitney *U* tests as appropriate by the distribution of the data. Linear regression was used to establish the correlation between costs and complications (measured by CCI), and the intersection of the line of best fit was used to determine at what baseline CCI NMP becomes profitable for a transplant program. All statistical analyses were performed using IBM SPSS Statistics V. 28.0. A *P*-value < 0.05 was considered statistically significant for all factors.

RESULTS

In total, 803 patients met the inclusion criteria. Of these, 174 (21.7%) were DCD grafts, and 629 (78.3%) were DBD; 174 cases (21.7%) were treated with end-ischemic NMP using OrganOx *Metra* through 9/30/2023 (Fig. 1A, B). Of these, 40 (22.9%) were DCD grafts, and 134 (77.1%) were DBD grafts. Before matching, the median UK-DCD score in NMP grafts was 7 points (IQR 5–8) versus 5 points (4–8, *P*=0.040) in the SCS cohort. In the DBD cohort, before matching, the median BAR score in NMP was 10 (4–13) and 9 (4–13) points in SCS (*P*=0.469). Demographics and donor and recipient risk factors for matched patients are provided (Table 1).

Overall Comparison of Complications and Costs

After matching, across all graft types, there was no difference in inpatient (NMP=27.6 vs 33.7 points, *P*=0.075) or cumulative 90-day CCI (39.1 vs 43.0 points, *P*=0.101) (Table 2). Organ acquisition/preservation was more expensive with NMP (\$88,923 vs \$52,446, *P*<0.001),

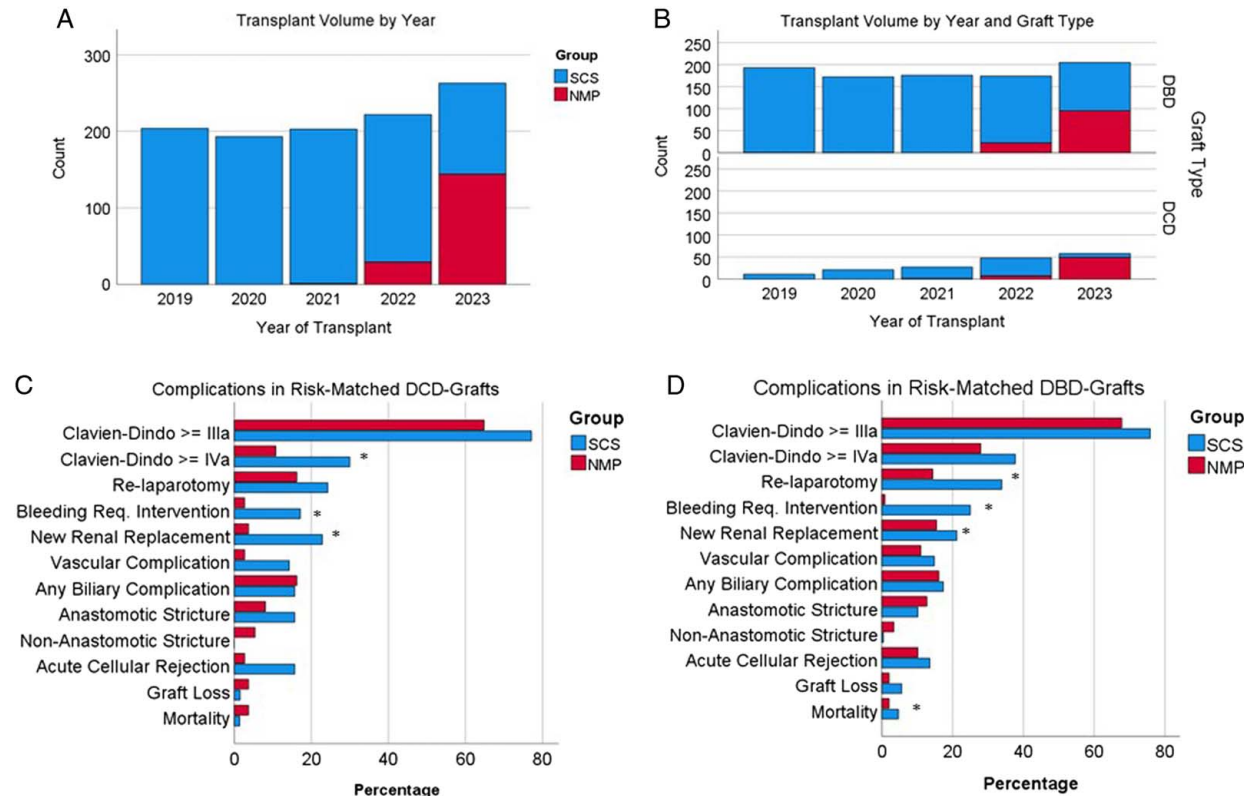


FIGURE 1. Transplant volumes and complications of transplant. A, Overall transplant volume by study year demonstrating increasing use of NMP (red) versus SCS (blue) throughout the period. B, Transplant volume by year split by graft type. C, Common liver transplant-associated complications for the risk-matched DCD cohort split by use of NMP (n = 37, red) versus SCS (n = 74, blue). D, Common liver transplant-associated complications for the risk-matched DBD cohort split by use of NMP (n = 118, red) versus SCS (n = 236, blue). *Denotes significance to $P < 0.05$ using χ^2 test. *Bleeding req. intervention refers to bleeding requiring procedural/surgical intervention (Clavien-Dindo $\geq$ IIIa. Nonanastomotic stricture refers to any biliary stricture superior/proximal to the biliary anastomosis.

yet cumulative costs were not different (\$221,952 vs \$219,049, $P = 0.726$) (Table 3). Sample costs for selected representative patients are available in Figure 2.

Including perfused and non-perfused grafts, the inpatient CCI was not different based on graft type (DCD = 27.6 vs DBD = 32.2, $P = 0.066$), though DCD-grafts did

TABLE 1. Postmatching Recipient and Donor Information and Transplant Risk Factors Based on the Use of Normothermic Machine Perfusion (NMP) Versus Static Cold Storage (SCS) in Donation After Brain Death (DBD) and After Cardiac Death (DCD) Grafts

	DCD-NMP (n = 37)	DCD-SCS (n = 74)	P	DBD-NMP (n = 118)	DBD-SCS (n = 236)	P
Bar Score ^{21,22} (DBD, points)	—	—	—	10 (4-13)	9 (5-14)	0.976
UK-DCD Score ²³ (DCD, points)	7 (5-8)	6 (4-8)	0.492	—	—	—
Recipient male sex	22 (59.4)	54 (67.6)	0.209	74 (62.7)	142 (60.1)	0.644
Recipient age (y)	64 (55-67)	61 (54-65)	0.731	60 (52-66)	60 (50-66)	0.696
Recipient BMI (kg/m ²)	27.3 (23.7-34.1)	28.0 (24.5-30.5)	0.974	28.1 (24.4-32.1)	27.9 (23.8-32.2)	0.642
Lab MELD at LT (points)	19 (13-21)	20 (12-29)	0.079	19 (14-27)	21 (13-29.3)	0.282
Donor age (y)	45 (30-55)	44 (32-55)	0.928	51 (38-62)	42 (29-55)	< 0.001
Donor BMI (kg/m ²)	24.1 (21.4-28.0)	29 (24-31)	0.083	28.1 (24.9-34)	27.4 (24-32)	0.220
Functional dWIT* (DCD, min)	20 (16-22)	23 (20-27)	0.051	—	—	—
Cold ischemia time† (h)	6.5 (5.6-7.1)	5.5 (5.1-6.4)	0.015	6.5 (5.5-7.7)	5.6 (4.5-6.5)	< 0.001
Total preservation time‡ (perfused cases, h)	16 (14-19)	—	—	14 (12-18.1)	—	—

Bold values are statically significant to $p < 0.05$.
 *Functional donor warm ischemia time is defined as time from agonal (in our center if SBP < 80 mm Hg or SaO₂ < 80%) to cold flush in the donor.
 †Cold ischemia time is defined as the time in hours from initiation of the flush solution to either initiation of NMP (in NMP-treated grafts) or to “liver out of ice” in SCS grafts.
 ‡Total preservation time is defined as the time from cross-clamp and flush in the donor to reperfusion in the recipient.
 dWIT indicates donor warm ischemia time; MELD, model for end stage liver disease.

TABLE 2. Risk-matched Complications and Outcomes by Use of Normothermic Machine Perfusion (NMP) Versus Static Cold Storage (SCS) in Donation After Brain Death (DBD) and After Cardiac Death (DCD) Grafts. Comparison to Published Benchmark Criteria Is Provided

	DCD benchmark ³⁰	DCD-NMP (n = 37)	DCD-SCS (n = 74)	P	DBD benchmark ²⁸	DBD-NMP (n = 118)	DBD-SCS (n = 236)	P
ICU LOS (d)	≤ 3	2 (2–4)	3 (2–6)	0.028	≤ 4	3 (2–4)	3 (2–4)	0.942
Hospital LOS (d)	≤ 16	11 (9–14)	14 (9–20)	0.183	≤ 18	15 (10–31)	17 (10–34)	0.253
Cumulative Inpatient CCI ²⁵ (points)	≤ 22.2	20.9 (0–34.8)	34.2 (8.7–47.5)	0.056	≤ 29.6	27.6 (12.9–44.8)	33.7 (8.7–54.2)	0.299
Cumulative 90-dCCI ²⁵ (points)	≤ 30.8	27.6 (20.9–44.3)	41.9 (27.1–54.3)	0.030	≤ 34.5	43.4 (27.6–58.1)	44.1 (26.22–63.6)	0.396
Clavien-Dindo ≥ IIIA ²⁴	≤ 56	24 (64.9)	54 (77.1)	0.174	≤ 54%	80 (67.8)	179 (75.8)	0.107
Clavien-Dindo ≥ IVa ²⁴	NA	4 (10.8)	21 (30.0)	0.026	NA	33 (28.0)	89 (37.7)	0.069
New renal replacement therapy, n (%)	≤ 9.6%	1 (2.7)	16 (22.9)	0.007	≤ 8%	15 (12.7)	50 (21.2)	< 0.001
Relaparotomy, n (%)	NA	6 (16.2)	17 (24.3)	0.334	NA	17 (14.4)	80 (33.9)	< 0.001
Vascular complication	NA	1 (2.7)	10 (14.3)	0.061	NA	13 (11.0)	35 (14.8)	0.323
Bleeding Req. intervention, n (%)	≤ 10.3	1 (2.7)	12 (17.1)	0.030	NA	1 (0.8)	59 (25.0)	< 0.001
Any biliary complication	NA	6 (16.2)	11 (15.7)	0.946	≤ 28	19 (16.1)	41 (17.4)	0.764
Nonanastomotic stricture, n (%)	≤ 8.8	2 (5.4)	1 (1.4)	0.051	NA	1 (0.4)	4 (3.4)	0.479
Anastomotic stricture, n (%)	≤ 16.8	3 (8.1)	11 (15.7)	0.267	NA	15 (12.7)	24 (10.2)	0.471
Acute Cellular Rejection (n/%)	NA	8 (21.6)	11 (15.7)	0.373	NA	17 (14.4)	26 (11.0)	0.357
90-day graft loss (n/%)	≤ 13.3%	1 (2.7)	1 (1.4%)	0.643	≤ 6	2 (1.7)	10 (4.2)	0.133
90-day mortality (n/%)	≤ 7.8%	1 (2.7)	1 (1.4%)	0.643	≤ 4	2 (1.7)	11 (4.7)	< 0.001

Bold values are statically significant to $p < 0.05$.

ITBL indicates ischemic type biliary session.

TABLE 3. Costs Associated With Use of Normothermic Machine Perfusion (NMP) Versus Static Cold Storage (SCS) in Donation After Brain Death (DBD) and After Cardiac Death (DCD) Grafts

	DCD-NMP (n = 37)	DCD-SCS (n = 74)	P	DBD-NMP (n = 118)	DBD-SCS (n = 236)	P
90-d before transplantation (pre-LT)	\$7,416 (\$2,864–\$12,173)	\$7878 (\$2,469–\$13,222)	0.923	\$3,926 (\$226–\$20,579)	\$12,683 (\$3,960–\$30,281)	< 0.001
Organ aquisition/ preservation	\$88,923 (\$88,923–\$88,923)	\$52,446 (\$52,446–\$61,958)	< 0.001	\$88,923 (\$88,923–\$88,923)	\$52,446 (\$46,079–61958)	< 0.001
Cost of transplant Admission* *Includes Organ Acquisition	\$168,650 (\$160,740–\$185,278)	\$155,980 (\$125,001–\$195,150)	0.284	\$197,939 (\$166,117–\$293,680)	\$164,318 (\$129,475–\$234,638)	< 0.001
Post-transplant admission cost	\$82,714 (\$76,449–\$98,052)	\$102,550 (\$73,972–\$142,151)	0.187	\$107,398 (\$84,056–\$206,102)	\$105,359 (\$73,651–\$175,313)	0.102
*Does not include organ acquisition						
Total 90-day pre- and post	\$194,225 (\$175,499–\$228,064)	\$194,185 (\$165,496–\$226,665)	0.932	\$234,552 (\$189,385–\$327,682)	\$218,443 (\$166,347–\$334,113)	0.267

Bold values are statically significant to $p < 0.05$.

The first * indicates (as noted in the table itself) that this cost includes organ acquisition. The second * indicates that this cost does not include that.

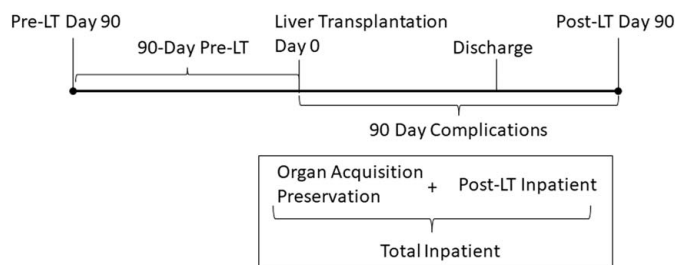


FIGURE 2. Timeline of included cost periods for cost analyses and cost breakdowns for selected patients with range of post-LT complications. Afib indicates atrial fibrillation; AKI, acute kidney injury; CRRT, continuous renal replacement therapy; DVT, deep vein thrombosis; NSTEMI, non-ST Elevation MI; PCI, percutaneous intervention; RVR, rapid ventricular response.

Study Groups	90 Day Complications				Costs			
Graft and Preservation	Highest Clavien Grade	Main Complication (Words)	90-Day CCI	90-day Pre-LT	Acquisition and Preservation	Post-LT Inpatient	Total Inpatient	Total 90-day Pre-/Post-LT
DCD-SCS	II	Hemolytic Anemia	24.2	\$11,428	\$52,446	\$103,005	\$155,451	\$175,592
DCD-NMP	II	AKI, Urinary Retention	20.9	\$4,566	\$88,923	\$68,299	\$157,221	\$178,860
DCD-SCS	IVa	AKI req. CRRT, Afib w RVR	60.2	\$1,060	\$52,446	\$147,492	\$199,937	\$216,960
DCD-NMP	IVa	NSTEMI req. PCI	47.3	\$849	\$61,958	\$84,959	\$146,917	\$214,848
DBD-SCS	II	CMV Viremia, DVT	36.2	\$19,938	\$50,615	\$71,415	\$122,030	\$262,410
DBD-NMP	II	Atrial Fibrillation, C Difficile	37.2	\$40,123	\$61,958	\$135,981	\$197,939	\$242,098
DBD-SCS	IVa	AKI req. CRRT, Aspiration PNA	68.7	\$16,884	\$50,615	\$150,872	\$201,487	\$406,208
DBD-NMP	IVa	AKI req. CRRT, Pericardial Effusion	73.2	\$20,724	\$88,923	\$160,528	\$249,450	\$313,945

demonstrate a lower median CCI at 90 days (37.2 vs 43.44, $P=0.014$). Organ Acquisition costs were not different between graft types (\$61,958 vs \$61,958, $P=0.309$). However, DCD-grafts did demonstrate lower inpatient (167,006 vs \$175,066, $P=0.047$) and lower cumulative costs (\$200,500 vs \$233,432, $P=0.001$) compared with DBDs.

DCD Cohort

Matching was achieved between 37 DCD-NMP and 74 DCD-SCS grafts. After matching, there was no difference in UK-DCD risk score (7 vs 6 points, $P=0.492$), recipient sex ($P=0.209$), recipient age (64 vs 61 years, $P=0.731$) or BMI (27.3 vs 28.0 kg/m², $P=0.974$), laboratory model for end stage liver disease (MELD) at LT (19 vs 20 points, $P=0.079$), donor age (45 vs 44 years, $P=0.928$) or BMI (24.1 vs 29.0 kg/m², $P=0.083$), or functional donor warm ischemia time (20 vs 23 minutes, $P=0.051$). There was a longer static cold storage (SCS)—time before perfusion in the NMP group after matching (6.5 vs 5.5 hours, $P=0.015$) [Table 1].

Complications

NMP-treated DCD-grafts experienced a reduction in the median CCI in the 90-day post-LT period (27.6, IQR 20.9–44.3) compared with the SCS cohort (41.9, 27.1–54.3, $P=0.030$), as well as a reduced ICU length-of-stay (LOS, 2 vs 3 days, $P=0.028$). They also experienced a reduction in number of Clavien grade $\geq$ IVa complications (10.8% vs 30.0%, $P=0.026$), post-LT RRT requirement (2.7% vs 22.9%, $P=0.007$), and bleeding requiring intervention (2.7% vs 17.1%, $P=0.030$). Remaining outcome measures, including the inpatient CCI (20.9 vs 34.2 points, $P=0.056$), 90-day graft loss (2.7% vs 1.4%, $P=0.643$), and 90-day mortality (2.7% vs 1.4%, $P=0.643$) were not different (Table 2, Fig. 1C, D). Perfused cases more similarly resemble previously established DCD benchmark values as appropriate for inpatient CCI (22.2 points), 90-day CCI (30.8 points), ICU LOS (≤ 3 days), bleeding ($\leq 10.3\%$), and renal replacement therapy ($\leq 9.6\%$).³⁰

Costs

The use of NMP was not associated with a difference in costs during the transplant admission, both including the cost of organ acquisition/preservation (\$168,650 vs \$155,980, $P=0.284$) and excluding that cost (\$82,714 vs \$102,550, $P=0.187$). These findings are despite a median increased cost of organ acquisition/preservation of \$36,447 with NMP (\$88,923 vs \$52,446, $P<0.001$). There was also no difference in cost in the 90-day pre-/post-LT period (\$194,225 vs \$194,185, $P=0.932$) or in the 90-day pre-LT period (\$7,416 vs \$7,878, $P=0.923$) [Table 3, Fig. 3A]. The costs of transplant by year are depicted in Figure 4A.

Recipients experiencing a Clavien-Dindo $\geq$ grade IVa complication had higher health care costs within 90 days following transplant ($P<0.001$) and in 90 days pre-/post-LT ($P<0.001$). Costs were correlated with increasing CCI using the equation Cost (DCD-NMP) = $2.03 \times 10^3 \times \text{CCI} + 1.47 \times 10^5$ (R^2 was 0.367). For SCS grafts, Cost (DCD-SCS) = $3.21 \times 10^3 \times \text{CCI} + 1.0 \times 10^5$ with an $R^2=0.340$. This calculation allowed us to identify a CCI threshold of 39.83 points in SCS-treated grafts when NMP becomes profitable in DCD grafts (Fig. 3C). The median CCI per study year is shown in Fig. 4C, demonstrating a reduction in CCI since the advent of NMP in 2022.

DBD -Grafts

Matching was achieved between 118 DBD-NMP and 236 DBD-SCS grafts. There was no difference between groups based on BAR score (10 vs 9, $P=0.976$), recipient sex ($P=0.644$), recipient age (60 vs 60 years, $P=0.696$), or BMI (28.1 vs 27.9 kg/m², $P=0.642$), laboratory MELD at LT (19 vs 21 points, $P=0.282$), or Donor BMI (28.1 vs 27.4 kg/m², $P=0.220$). NMP-treated grafts were at higher risk with respect to donor age (51 vs 42 years, $P<0.001$) and CIT (6.5 vs 5.6 hours, $P<0.001$) [Table 1].

Complications

There was no difference in CCI for DBD-grafts based on NMP for the inpatient (27.6 vs 33.7, $P=0.299$) or 90-day periods (43.4 vs 44.1, $P=0.396$). NMP treatment was associated with a reduced rate of post-LT RRT (12.7% vs

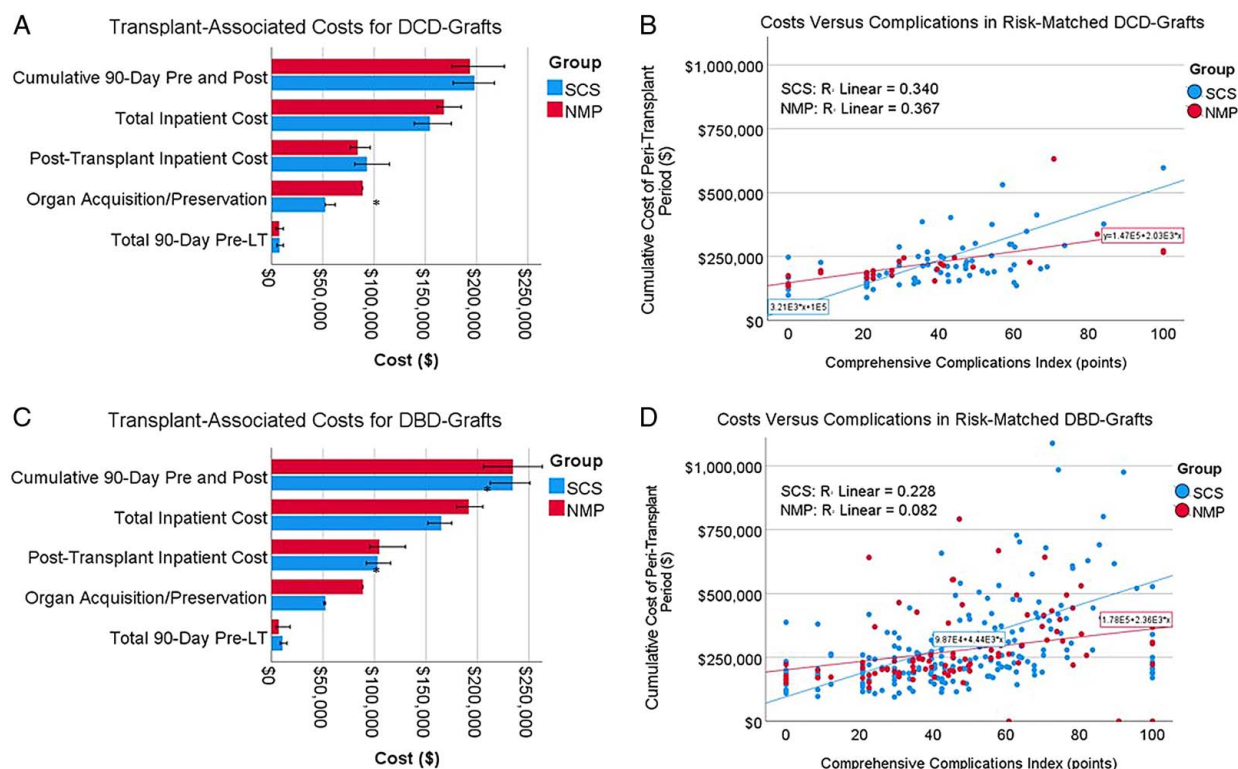


FIGURE 3. Costs of liver transplantation in the 90 days following transplantation with Normothermic Machine Perfusion (NMP) versus the traditional Static Cold Storage (SCS) using Risk-Adjusted Matching. A, Actual transplant-associated costs in matched DCD-grafts. B, Correlation between peritransplant costs and comprehensive complications index (CCI) in DCD-grafts demonstrating relative profitability of NMP when CCI is at least 39.8 points. C, Actual transplant-associated costs in matched DBD grafts. D, Correlation between peritransplant costs and CCI in DBD-grafts demonstrating relative profitability of NMP when CCI is at least 38.13 points. *Denotes significance to $P < 0.05$ using Mann-Whitney U test. Note that post-LT inpatient cost reflects the cost from the conclusion of transplantation to discharge, while total inpatient cost includes surgical costs and the costs of organ acquisition/preservation.

21.2%, $P < 0.001$), relaparotomy (14.4% vs 33.9%, $P < 0.001$), and bleeding requiring reintervention (0.8% vs 25.0%, $P < 0.001$). Interestingly, there was a reduced rate of 90-day mortality with NMP (1.7% vs 4.7%, $P < 0.001$) but no change in the rates of graft loss (1.7% vs 4.2%, $P = 0.133$) or other complications (Table 2, Fig. 1). NMP-cases did more similarly mimic benchmark cases for inpatient CCI (≤ 29.6) but not 90-day CCI (≤ 34.6), as well as post-op RRT ($\leq 8\%$).²⁸

Costs

While NMP was associated with an increased cost of the transplant admission (\$197,939 vs \$164,318, $P < 0.001$), there was no difference in the total 90-day pre-LT and post-LT costs (\$234,552 vs \$218,443, $P = 0.267$). A reduction in the 90-day costs pre-LT (\$3,926 vs \$12,683, $P < 0.001$) was seen. As expected, there was a similar increase in the costs of organ acquisition/preservation with NMP (\$88,923, IQR \$88,923–\$88,923) versus SCS (\$52,446, \$46,079–\$61,958) (Table 3, Fig. 3C). The median transplant costs by year are depicted in Figure 4B.

Recipients experiencing a Clavien-Dindo $\geq IVa$ complication had higher health care costs in the 90 days following transplant ($P < 0.001$) and the 90 days pre-/post-LT ($P < 0.001$). Costs were correlated with increasing CCI using the equation Cost (DBD-NMP) = $\$2.36 \times 10^3 \times \text{CCI} + \1.78×10^5 . R^2 for this correlation was 0.186. For SCS grafts, Cost (DBD-SCS) = $4.44 \times 10^3 \times \text{CCI} + \9.87×10^4

with $R^2 = 0.235$. This allows us to identify a CCI for SCS grafts = 38.125 at which point NMP becomes profitable in DBD grafts (Fig. 3D). The median CCI per study year is shown in Figure 4D, again demonstrating a reduction in CCI since the advent of NMP in 2022.

DISCUSSION

This is the first work to simultaneously compare the complications and costs of LT in grafts receiving end-ischemic, back-to-base NMP versus their counterparts treated with SCS and the first to do so using a systematic, risk-matched approach. We first find that NMP is associated with a reduction in cumulative 90-day complications in DCD grafts, with less postoperative renal replacement therapy and serious bleeding events. Second, in DBD grafts, NMP was associated with reductions in bleeding events, relaparotomy, and 90-day mortality, though the cumulative total complications were not different. Third, regarding costs, NMP cases did not demonstrate an increase in costs for the 90-day pretransplant and post-transplant period (DBD and DCD grafts), or an increase in transplant admission costs (DCD-grafts), despite nearly \$40,000 additional costs for organ acquisition/preservation. This points to other cost-savings associated with earlier transplantation, reduced early complications, and/or improved logistics that offset the up-front costs of NMP. Finally, we also find a

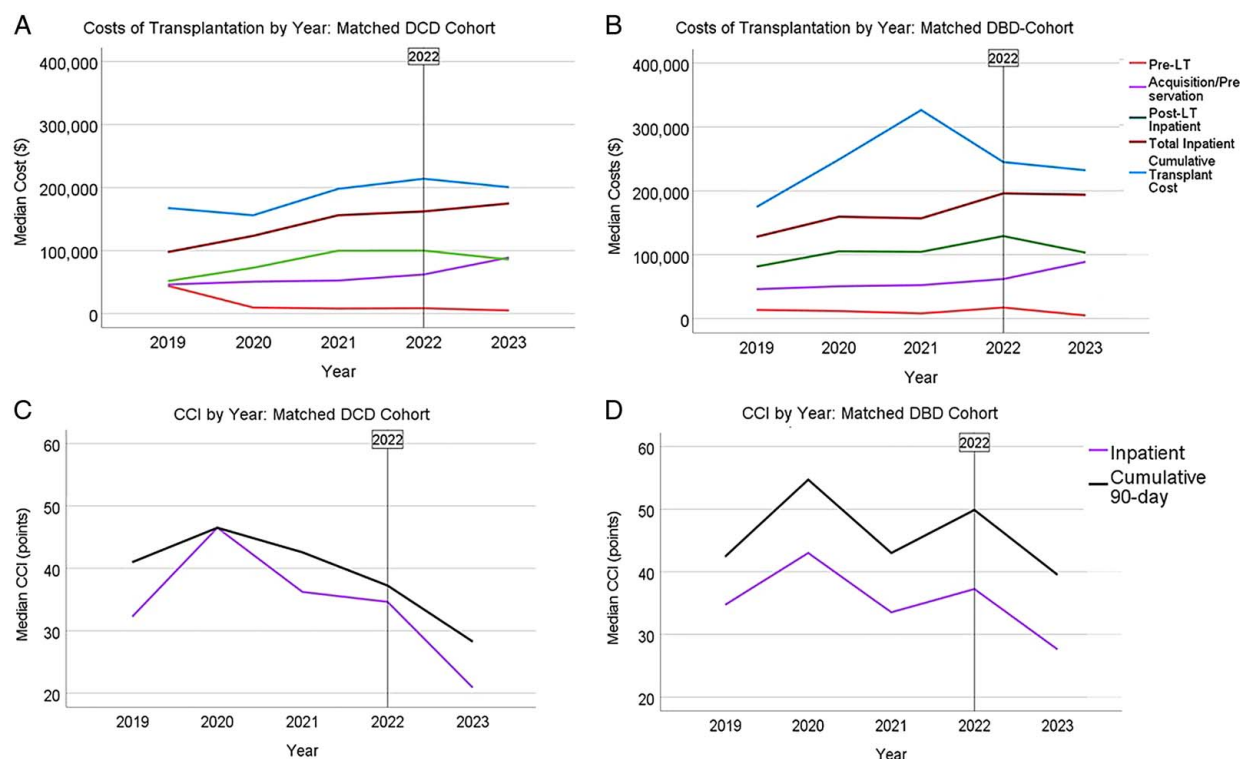


FIGURE 4. Costs and complications of liver transplantation by study year. A, Costs associated with transplantation in donation after cardiac death (DCD) grafts by year of transplantation, including all grafts in risk-matched cohort regardless of the use of NMP. B, Costs associated with donation after brain death (DBD) grafts. C, Median Comprehensive Complications Index (CCI) for DCD grafts for the inpatient stay (from LT to discharge) and for the cumulative 90-day post-transplant period. Includes all grafts in risk-matched cohort regardless of the use of NMP. D, Median CCI in DBD grafts.

reduction in complications and costs with DCD grafts, generally supporting an increased use of these grafts.

Since the approval of NMP by the food and drug administration in 2021, there has been a rapid change in the clinical practice of liver preservation in the United States.³¹ However, both the financial and clinical impacts of this change remain relatively under-explored at this time. Preliminary data from US and Canadian centers have shown improved early graft function in DCD grafts, and one study demonstrated improved 30-day graft survival with NMP.^{16,32} Studies at our center and others have also shown the ability to use livers declined by others, thereby increasing liver utilization.^{19,33–35} A recent international study on the utilization of controlled DCD livers demonstrated improved liver utilization when different machine perfusion concepts were applied.³⁶ This risk-matched analysis demonstrates some clear benefits of NMP in both DBD- and DCD-cohorts. Of note is the reduction in cumulative complications in DCD grafts and a notable reduction in 90-day mortality in the DBD cohort. Median CCI and complication rates are slightly higher in our cohort than established benchmarks in both DCD and DBD-grafts, though recipient risk measured by MELD was above benchmark values in our cohort.^{28,30} This may be explained by the inclusion of non-benchmark cases, most notably DCD and DBD recipients with chemical MELD > 20 points, and median BAR score of 10&9 points in our cohort versus 4 points in DBD benchmark cases, though this cannot be stated definitively. We do note that perfused cases more closely resemble benchmark outcomes despite minimal

change in baseline risk, providing some evidence that NMP can help achieve ideal outcomes even in higher-risk cases. The CCI has been well linked to outcomes of LT, including by Lai et al,³⁷ who showed a strong correlation between CCI and graft outcomes (90-day and 1-year) and overall survival. Both graft types also experienced a reduction in post-LT need for renal replacement therapy in our study here. Although NMP triggers ischemia-reperfusion injury (IRI) on the device, this perfusion technique reduces the additional static cold storage.³⁸ The IRI during NMP induces cytokine and Damps molecule release into perfusates, together with liver transaminases, which become discarded at the end of NMP.³⁹ Despite the activation of liver tissue, the number of proinflammatory molecules immediately released into the recipient's circulation is lower compared to SCS grafts with a subsequent reduction of systemic inflammation in the recipient.⁴⁰ The observed impact on immediate clinical outcome parameters is the direct consequence, including less systemic hypoperfusion and renal-protective effect compared to its SCS counterpart.^{41–43} NMP is also known to reduce cytokine concentration in the transplanted graft and reduce inflammatory mediators, both due to transcriptional changes and the mechanical act of flushing these cytokines into the perfusate, which is then discarded.⁴⁴ It is thus reasonable to think that our findings of reduction in renal complications and less serious bleeding events may result from a reduction in early systemic inflammation with NMP.

While improving outcomes should be our primary aim, knowledge of the relative costs of various health care

resources is critical to the proper allocation of resources. This study is, to our knowledge, the first to examine the risk-adjusted costs of NMP, including both costs specific to preservation and across the total transplant period. More specifically, we report costs associated with a back-to-base model of machine perfusion, which is logistically less challenging than a device-to-donor approach. Webb et al⁴⁵ previously demonstrated a theoretically similar cost of NMP versus SCS analysis using a Markov model, though this was not an actual reporting of cost data. The same group also demonstrated an actual cost in Canadian dollars of \$35,659–\$48,076 per NMP-run, depending on the location of procurement.¹⁸ A US-specific Markov analysis predicted an increased cost of NMP versus SCS using Scientific Registry of Transplant Recipients data, but, again, this was a model rather than true data.⁴⁶ We report that, while NMP was more expensive with respect to organ acquisition and preservation, there was no difference in the costs of the 90-day pre- and post-transplant period. DCD-grafts also demonstrated no difference in the cost of the transplant admission, and though DBD transplantations were more expensive, cost savings in the pretransplant period were able to offset this in the global period. In addition, we find that NMP becomes cost-effective at a baseline CCI of just under 40 points, which is less than the median CCI in both SCS cohorts, explaining why NMP might be, on average, cost-effective in the population at large. In addition to the described clinical benefits, NMP offers logistical benefits also, including improved provider quality-of-life and the ability to balance heavy caseloads at various times of the day.^{31,47} These findings have also been confirmed in hypothermic perfusion, such as by Brüggewirth et al,⁴⁸ who demonstrated the logistical benefits of HOPE without increasing clinical risk. Previous studies, such as by De Simone and Ghinolfi,⁴⁹ have cited complications-based benefits but offer concerning findings regarding the cost-effectiveness of NMP. Indeed, Zimmerman and Carter⁵⁰ in 2022 published a UK-based cost model raising concern that machine perfusion might never become cost-effective. Finally, the 2018 ASTS White paper on the topic predicted cost increases up to \$50,000 per device, listing this as a likely barrier to implementation.⁵¹

However, risk-matched cases employing NMP in our study are, at worst, equally costly to SCS cases, indicating that surgeons may feel comfortable using NMP, despite its' up-front cost, without wasting valuable hospital financial resources. This is in line with the response of Boteon et al to the Zimmerman study, predicting that complication-reduction and increasing NMP volume may help improve the overall cost landscape.⁵² Previous works have estimated end-ischemic NMP to cost USD \$40k/€10k (in US/Europe) and hypothermic-oxygenated perfusion (HOPE) is €3.5–6.5k in Europe and estimated to be \$15 to 20k once approved in the United States. Our findings support similar per-use costs of NMP, but note this does not translate to overall increases in costs.⁵³ Previous work has demonstrated increased costs in DCD grafts, though Muller and colleagues have demonstrated the ability of HOPE to mitigate cost increases associated with DCDs.^{54,55} Thus, cost-savings may increase further when the less-expensive HOPE is introduced to the US market. We do acknowledge that our findings are likely more reproducible in larger programs secondary to the necessary infrastructure, training, and overall machinery needed, in a manner similar to that described by Hunt et al⁵⁶

with normothermic regional perfusion (NRP) programs. Webb et al⁴⁵ further demonstrated incremental improvement in Quality-Associated-Life-Years (QALYs) of over 5 years with NMP, which correlates to expected cost savings of \$50,000 in this model. It is reasonable to think these cost-savings may also be seen in pre-transplant care, as previous studies estimate monthly spending as high as \$30,000 (MELD $\geq$ 30) or \$37,000 (MELD $\geq$ 35) in patients listed for LT in certain regions.⁵⁷

Previous studies looking at outcomes of NMP have not found a protective effect with respect to biliary strictures.^{16,58} Indeed, works by Watson and colleagues have both demonstrated a persistently high risk of non-anastomotic strictures with end-ischemic NMP, highlighting a remaining concern of this approach.^{16,33,58} Gilbo and colleagues paralleled this concern demonstrating higher biliary injury scores with NMP in human livers using grafts from the COPE consortium RCT published in 2018 by Nasralla et al.^{59,60} Unfortunately, while NMP was protective for certain complications in our cohort, we did not find a beneficial effect of NMP on overall or specific biliary complications, including anastomotic or nonanastomotic strictures within our early 90-day timeframe after transplantation. We do note that CIT was significantly higher in the NMP cohort, which could bias these results, but the impact of this cannot be stated definitively. While not yet approved for routine use in the United States, HOPE has demonstrated impressive protective effects on the biliary system in US-based clinical cohort and animal studies and, indeed, in RCTs and large cohort studies in European countries, where this preservation technique is increasingly commissioned for DCD liver preservation.^{10,53,61–64} This was most notably shown in the 2021 Van Rijn trial, a randomized trial demonstrating a reduction in the number and severity of intrahepatic nonanastomotic biliary strictures in HOPE-treated DCD-grafts.¹⁰ Real-world data from Italy have paralleled these findings in extended DBD and DCD livers.^{65,66} While our study does demonstrate clear benefits to NMP, protection of the biliary tree remains a concern, and future works should systematically compare HOPE and NMP, as well as establish improved markers for viability assessment that are unique to the biliary tree.

One of the most relevant benefits of perfusion techniques is viability assessment during perfusion, essential for a safe increase of the used ECD and particularly DCD livers, for which various approaches have been described without a clear current consensus.^{67,68} This is particularly important as we find that we are using a higher rate of DCD grafts with the NMP program. Interestingly, we observe DCD grafts to have less complications and reduced costs compared to their DBD counterparts, though this is likely limited by selection bias as surgeons are prone to have higher standards for graft-related risk in DCD grafts. Usually, we conduct a general macroscopic liver assessment (visually at the donor site, before and after flush) along with biochemical perfusate and bile analysis, including assessment of lactate levels, pH, glucose metabolism, and analysis of bile flow and biochemistry.^{19,33,69} Viability assessment was also recently developed and validated for HOPE in livers. The group from Zurich has described the key role of mitochondrial markers representing function and injury of complex I and its release during reperfusion, namely showing that perfusate levels of Flavin mononucleotide (FMN) and nicotinamide adenine dinucleotide can be

quantified in real-time using spectroscopy.^{13,70,71} These can be combined with conventional markers such as lactate, and surgeons at our centers do routinely discard marginal organs if such markers are concerning.^{13,67,68,72,73} Perfusate FMN was linked to post-transplant graft loss, graft function, and complications, and the role of FMN in other organs was recently described.^{74–76} While perfusate thresholds during HOPE before LT are established and validated, the role of this marker during NMP and other perfusion techniques is currently being explored. Overall, the final decision on organ viability for transplant remains to the judgment of the operating surgeon at this time, though ongoing research does continue to elucidate what criteria are most helpful in this assessment.

Limitations to this study include the retrospective nature of the work. We performed risk-matching using well-validated risk scores in an attempt to create equal risk groups for comparison, but this cannot fully remove the potential for confounding variables. Outcomes such as CCI are multifactorial and can be influenced by many parameters that are irrespective of even standard graft/recipient risk models. Further, we matched using 2 risk scores, both of which include CIT as a variable, yet there remained a longer CIT in the NMP cohort in both groups. This could be viewed as making complication-reduction more impressive, as they are found despite slightly longer CIT. NMP does require time to connect the liver to the device, which may contribute to a necessary increase in CIT. However, we acknowledge that this does introduce bias into our between-group comparison. Because our providers feel more comfortable with a little longer CIT with NMP, matching for these variables independently (rather than using risk score) would have introduced severe limitations to our sample size, thus we elected for our approach. Comparison of our findings to benchmark values is challenging. While most exclusions are shared between studies (retransplantation, split grafts, combined), our study does include patients with MELD ≥ 20 , and the median BAR score in our cohort was 10 and 9 points versus 4 in the relevant benchmark study.^{28,30} Thus, we might expect a greater complication rate, as we do, in fact, see. In addition, costs of transplantation are also multifactorial and thus may be influenced by nongraft-related events. Further, we cannot ascertain the true hospital system effects of NMP, such as changes in OR personnel salary associated with less overnight transplants, as these are typically not captured in the patient-associated costs we were able to obtain. Only costs from our own health care system are included, which means that these costs likely underestimate the total health care costs a patient incurred. We suspect this disproportionately affects pre-LT costs, as those costs have been estimated as higher in previous studies,^{57,77} and patients typically receive care from more local institutions before their transplant versus after when care is centered around the transplant center itself. This study specifically reports the outcomes and costs of the “back-to-base” approach to NMP, and thus the more costly and challenging “device-to-donor” approach may or may not demonstrate similar feasibility. Finally, dry run costs, which are typically higher in DCD grafts, were not amortized over the DCD cohort, as they could not be accurately included when risk-matching is performed. However, as cited by Wall et al,⁵⁵ it is likely that these would increase overall costs in DCD-grafts, and future studies should incorporate these costs into perfusion-focused analyses.

CONCLUSIONS

NMP was associated with improvements in certain early complications in this risk-matched comparison to SCS controls, including post-LT dialysis, serious bleeding events, and a reduced cumulative CCI in DCD grafts. These improvements came without an increase in total peri-transplant costs, despite the initial jump in costs associated with organ preservation and acquisition. Providers could comfortably use NMP in the United States in the context of their overall health care costs in their own program, though future studies should clinically compare the impact of normothermic and hypothermic perfusion techniques when available.

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